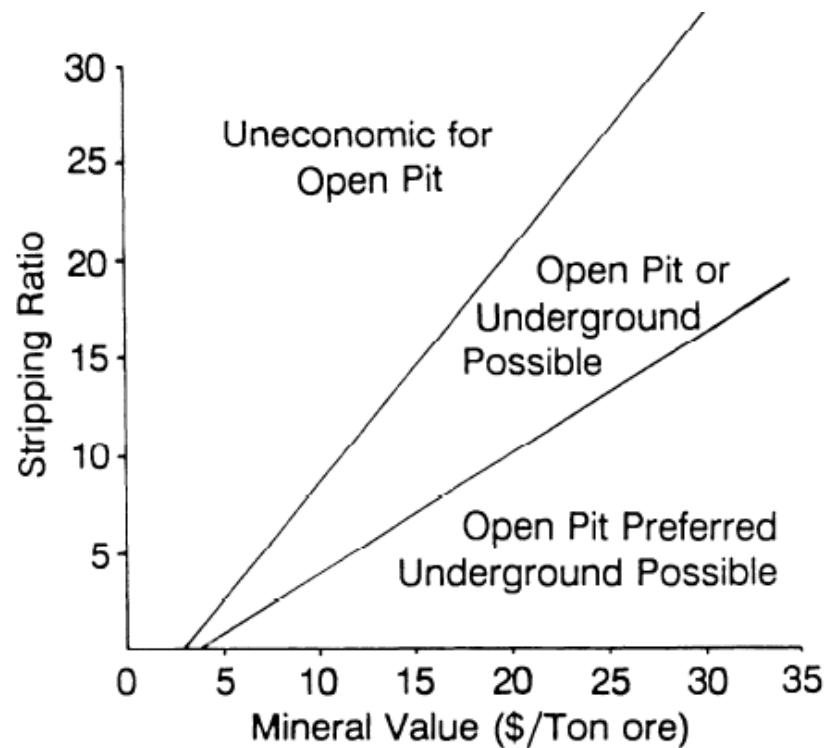


Mining method selection techniques



The way to mine ore deposit

surface, underground, Novel

- ▶ Surface mining
 - ▶ Open pit
 - ▶ Open cast/ strip
 - ▶ Placer mining
 - ▶ Solution mining

- ▶ Underground mining
 - ▶ BORD AND PILLAR
 - ▶ BLOCK CAVING
 - ▶ LONGWALL/SHORTWALL MINING
 - ▶ ROOM AND PILLAR/STOPE AND PILLAR MINING
 - ▶ SUBLEVEL STOPING
 - ▶ SUBLEVEL CAVING
 - ▶ SHRINKAGE STOPING
 - ▶ CUT AND FILL STOPING
 - ▶ SQUARE SET STOPING
 - ▶ TOP SLICING

NOVEL MINING METHODS

- ▶ Rapid excavation.
- ▶ Automation and robotics.
- ▶ Hydraulic mining.
- ▶ Methane drainage.
- ▶ Underground gasification and combustion.
- ▶ Marine mining.
- ▶ Nuclear mining.

Selection based on

- ▶ Physical and geologic characteristics of the deposit.
- ▶ Ground conditions of the hanging wall, footwall, and ore zone.
- ▶ Mining and capital costs.
- ▶ Mining rate.
- ▶ Availability and cost of labour.
- ▶ Environmental considerations.

There is no single appropriate mining method for a deposit; there are usually two or more feasible methods. Each method entails some inherent problems. Consequently, the optimum method is that method with the least problems.

The aim of determining the appropriate mining method is

**maximize company profits,
maximize recovery of the mineral, and
provide a safe environment for the miners**

None of the method selection systems deal with in situ stress. The techniques do account for the vertical stress via depth, but none of the methods discuss how a high horizontal stress impacts the choice of the mining method.

SELECTION METHOD TECHNIQUES

Boshkov and Wright

- ▶ one of the first qualitative classification schemes developed for underground method selection.
- ▶ Their system assumes that the possibility of surface and underground mining has already been eliminated.
- ▶ It uses general descriptions of the ore thickness, ore dip, strength of the ore, and strength of the walls to identify common methods that have been applied in similar conditions.
- ▶ The results of this classification provide up to four methods that may be applicable.

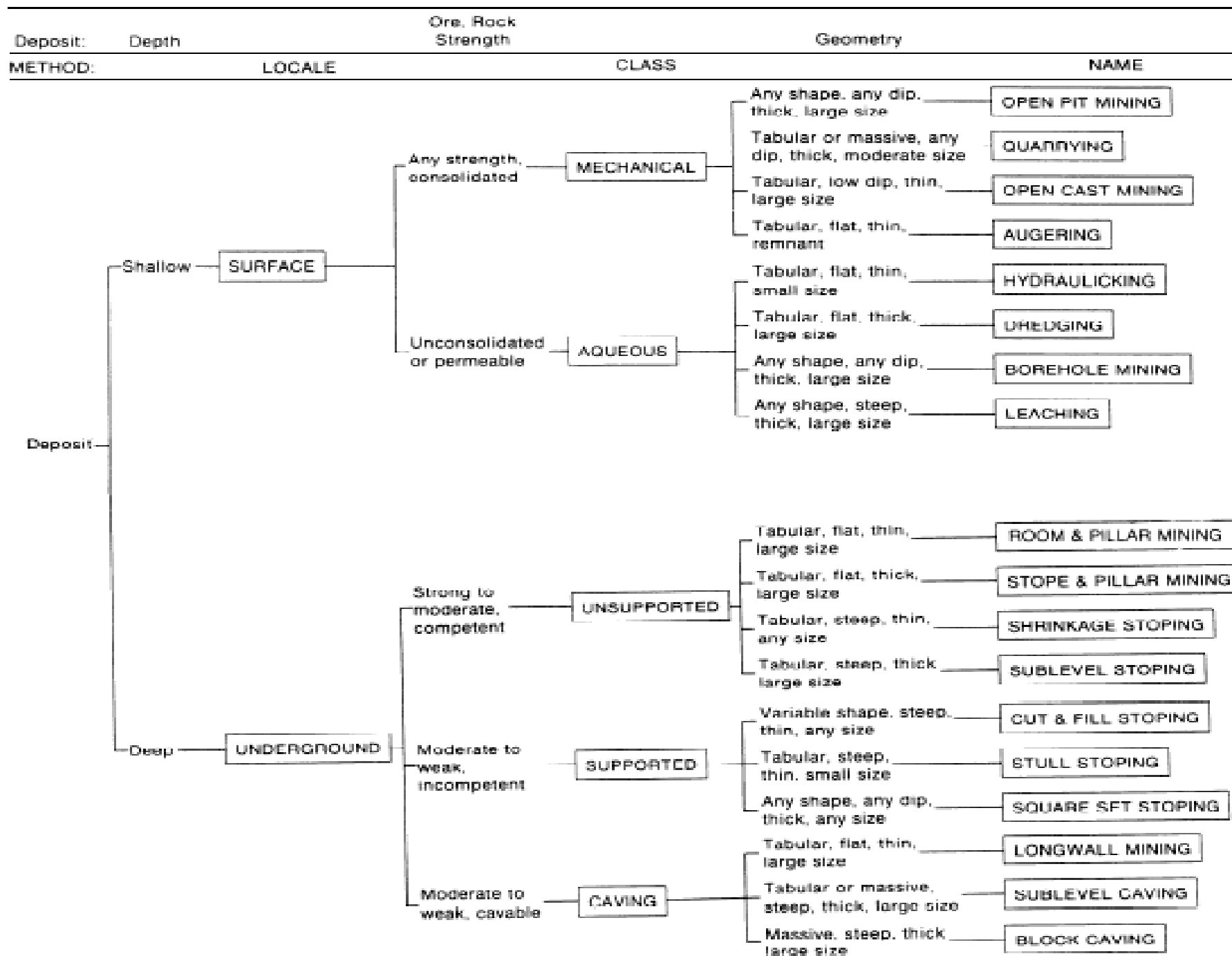
Type of Ore Body	Dip	Strength of Ore	Strength of Walls	Commonly Applied Methods of Mining
Thin beds	Flt	Stg	Stg	Open stopes with casual pillars Room and pillar Longwall
		Wk or Stg	Wk	Longwall
Thick beds	Flt	Stg	Stg	Open stopes with casual pillars Room and pillar
		Wk or Stg	Wk	Top slicing Sublevel caving
		Wk or Stg	Stg	Underground glory hole
Very thick beds				Same as for masses
Very narrow veins	Stp	Stg or Wk	Stg or Wk	Resuing
Narrow veins (widths up to economic length of stull)	Flt			Same as for thin beds
	Stp	Stg	Stg	Open stopes Shrinkage stopes Cut and fill stopes
			Wk	Cut and fill stopes Square set stopes
			Stg	Open underhand stopes Square set stopes
			Wk	Top slicing Square set stopes

Wide veins	Flt Stp	Stg	Stg	Same as for thick beds or masses Open underhand stopes Underground glory hole Shrinkage stopes Sublevel stoping Cut and fill stopes Combined methods
			Wk	Cut and fill stopes Top slicing Sublevel caving Square set stope Combined methods
			Wk	Open underhand stopes Top slicing Sublevel caving Block caving Square set stopes Combined methods
			Wk	Top slicing Sublevel caving Square set stopes Combined methods
			Stg	Underground glory hole Shrinkage stopes Sublevel stoping Cut and fill Combined methods
			Wk or Stg	Top slicing Sublevel caving Block caving Square set stopes Combined methods
Masses		Stg	Stg	Underground glory hole Shrinkage stopes Sublevel stoping Cut and fill Combined methods
		Wk	Wk or Stg	Top slicing Sublevel caving Block caving Square set stopes Combined methods

Wk = weak; stg = strong; flt = flat; stp = steep.

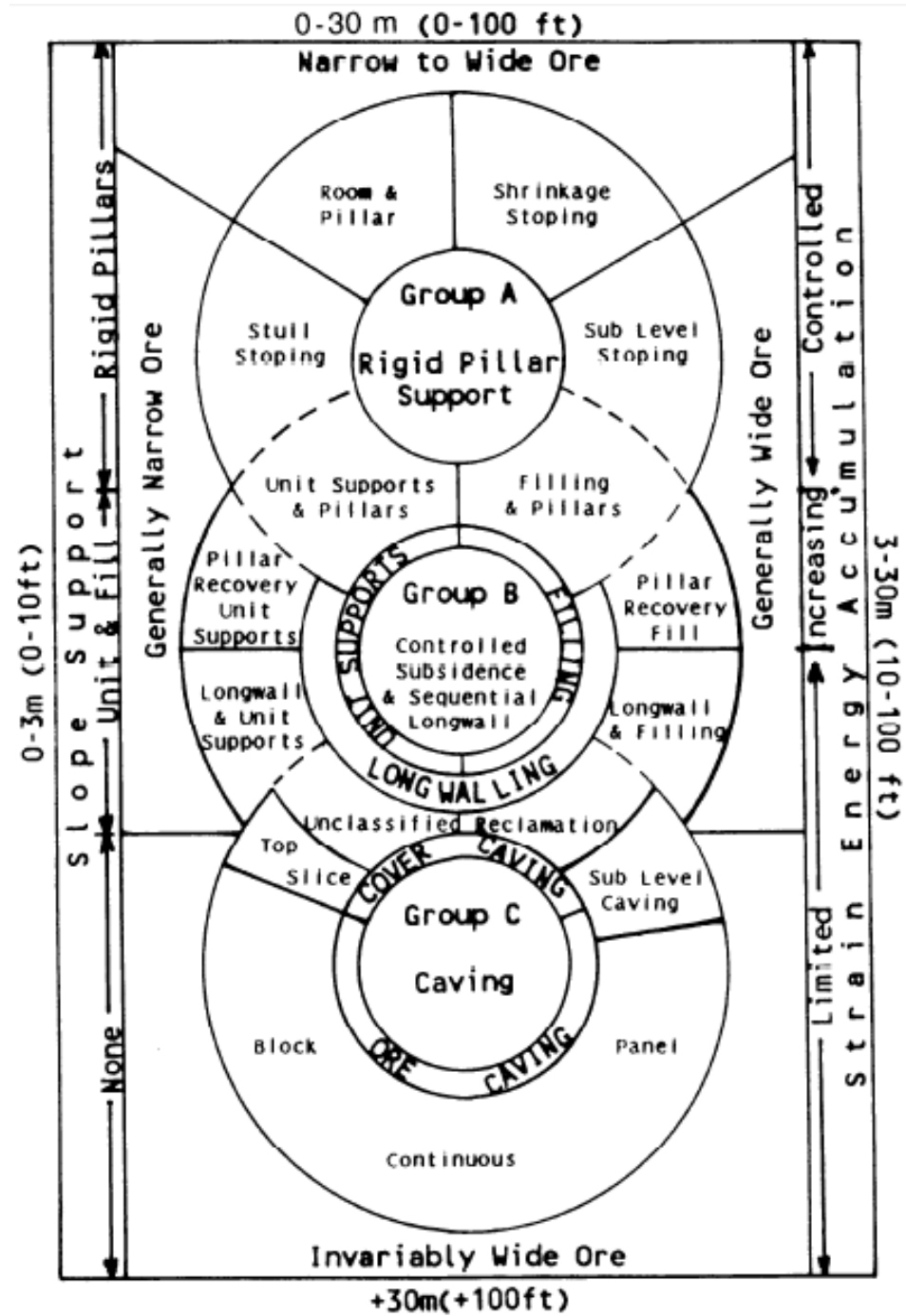
Hartman

- ▶ Developed a flow chart selection process for defining the mining method, based on the geometry of the deposit and the ground conditions of the ore zone
- ▶ This classification includes surface and underground methods, coal, and hard rock



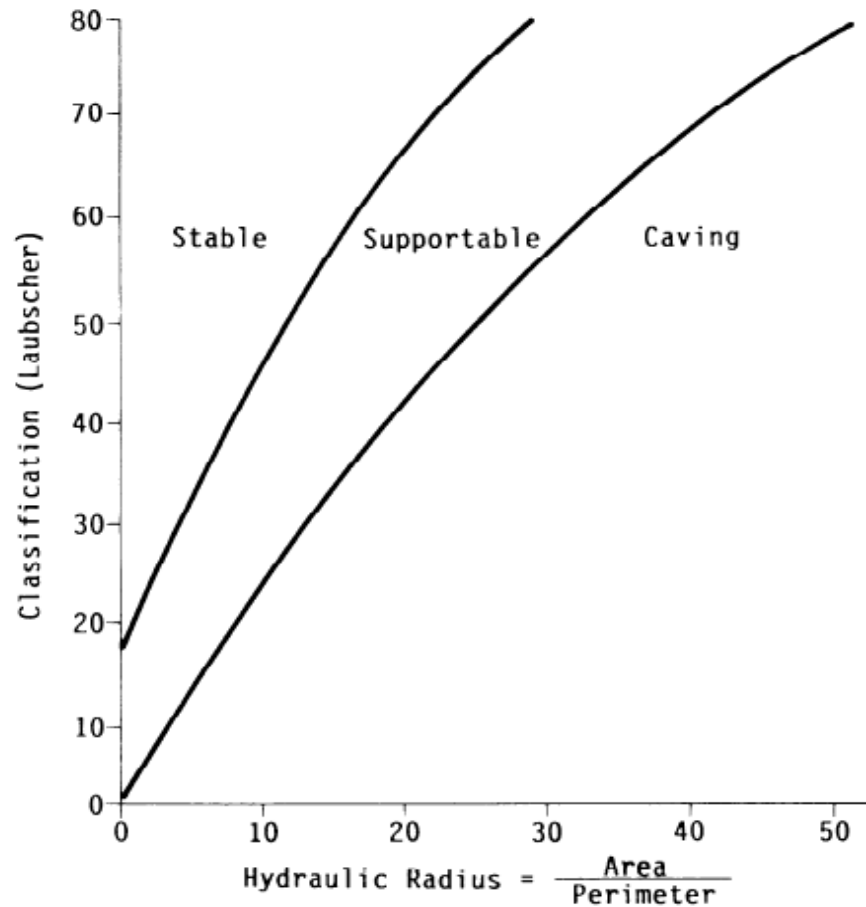
Morrison

- ▶ The classification system proposed by Morrison (1976) divides underground mining into three basic groups: (1) rigid pillar support, (2) controlled subsidence, and (3) caving
- ▶ General definitions of ore width, support type, and strain energy accumulation are used as the criteria for determining a mining method.
- ▶ This classification helps to demonstrate the selection continuum, choosing one method over another based on the various combinations of ground conditions.
- ▶ In this system, the ground conditions have already been evaluated to determine the type of support required.

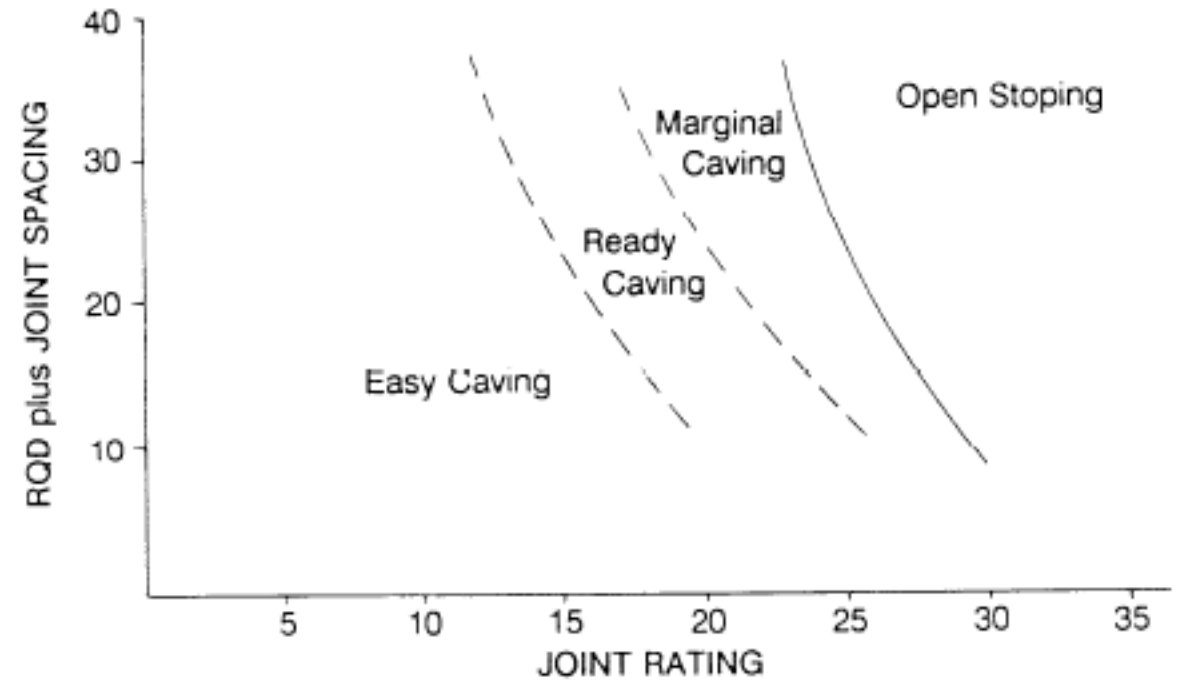


Laubscher

- ▶ The selection process is based on his **rock mass classification system**, which adjusts for expected mining effects on the rock mass strength
- ▶ Laubscher's scheme is aimed at the **mass mining methods**, primarily block caving vs stoping; his main emphasis is on cavability.
- ▶ The parameters that determine whether a caving system is used over a stoping system are the degree of fracturing, RQD (rock quality designation), joint spacing, and the joint rating, which is a description of the character of the joint, i.e. filling, and water conditions
- ▶ This scheme puts emphasis on the jointing as the only control for determining cavability
- ▶ Laubscher (1990) has recently modified the classification to relate his rock mass rating to the **hydraulic radius**.
- ▶ **By including the hydraulic radius, cavability becomes feasible for more competent rock if the area available for undercutting is large.**



Laubscher's cavability based on hydraulic radius and classification



Laubscher's classification for cavability evaluation

Nicholas

- ▶ The classification determines feasible mining methods by **numerical ranking and thus is truly quantitative**
- ▶ **Numeral values varies from -49 to 4**
- ▶ The first step is to classify the **ore geometry and grade distribution**
- ▶ The rock mechanics characteristics of the ore zone, hanging wall, and footwall are similarly classified
- ▶ A numerical ranking is then performed by adding up the values of each mining method
- ▶ The net weighting is then multiplied by each of the categories.
Those two or three mining methods that have the highest overall numerical ranking should be economically analyzed.

Definition of Deposit Geometry and Grade Distribution

- 1) *General Shape/Width*
 - equi-dimensional
 - all dimensions are on the same order of magnitude
 - platy—tabular
 - two dimensions are many times the thickness, which does not usually exceed 325 ft (100 m)
 - irregular
 - dimensions vary over short distances
- 2) *Ore Thickness*
 - narrow
 - < 30 ft (< 10 m)
 - intermediate
 - 30–100 ft (10–30 m)
 - thick
 - 100–325 ft (30–100 m)
 - very thick
 - > 325 ft (> 100 m)
- 3) *Plunge*
 - flat
 - < 20°
 - intermediate
 - 20°–55°
 - steep
 - > 55°
- 4) *Depth Below Surface*
 - provide actual depth
- 5) *Grade Distribution*
 - Uniform.* The grade at any point in the deposit does not vary significantly from the mean grade for that deposit.
 - Gradational.* Grade values have zonal characteristics, and the grades change gradually from one to another.
 - Erratic.* Grade values change radically over short distances and do not exhibit any discernible pattern in their changes.

Rock Mechanics Characteristics

1) *Rock Substance Strength*

(uniaxial strength/overburden pressure)

weak	< 8
moderate	8–15
strong	> 15

2) *Fracture Frequency*

	No. of Frac- tures per		% RQD
	(ft)	(m)	
very close	> 5	> 16	0– 20
close	3–5	10–16	20– 40
wide	1–3	3–10	40– 70
very wide	< 1	< 3	70–100

3) *Fracture Shear Strength*

weak	clean joint with a smooth surface or fill with material with strength less than rock substance strength
moderate	clean joint with rough surface
strong	joint is filled with a material that is equal to or stronger than rock substance strength

Problem

Input data	Description
General Deposit Shape	tabular or platy
Plan width	330 ft (100 m)
Ore thickness	300 ft (90 m)
Ore plunge	15°
Grade distribution	primarily disseminated
Depth	1150 ft (350 m)
Ore Zone	
Rock substance strength	12,000 psi (83 MPa)
Fracture spacing	3 frac/ft (10 frac/m)
RQD	40
Fracture shear strength	clean joint with a rough surface
Hanging Wall	
Rock substance strength	16,000 psi (110 MPa)
Fracture spacing	1.2 frac/ft (4 frac/m)
RQD	65
Fracture shear strength	clean joint with a rough surface
Footwall	
Rock substance strength	7,000 psi (48 MPa)
Fracture spacing	4.3 frac/ft (14 frac/m)
RQD	40
Fracture shear strength	joints coated with thin clay

Table 23.4.4. Ranking of Geometry/Grade Distribution for Different Mining Methods

Mining Method	General Shape			Ore Thickness				Ore Plunge			Grade Distribution		
	M	T/P	I	N	I	T	VT	F	I	S	U	G	E
Open Pit Mining	3	2	3	2	3	4	4	3	3	4	3	3	3
Block Caving	4	2	0	-49	0	2	4	3	2	4	4	2	0
Sublevel Stoping	2	2	1	1	2	4	3	2	1	4	3	3	1
Sublevel Caving	3	4	1	-49	0	4	4	1	1	4	4	2	0
Longwall Mining	-49	4	-49	4	0	-49	-49	4	0	-49	4	2	0
Room and Pillar Mining	0	4	2	4	2	-49	-49	4	1	0	3	3	3
Shrinkage Stoping	2	2	1	1	2	4	3	2	1	4	3	2	1
Cut and Fill Stoping	0	4	2	4	4	0	0	0	3	4	3	3	3
Top Slicing	3	3	0	-49	0	3	4	4	1	2	4	2	0
Square Set Stoping	0	2	4	4	4	1	1	2	3	3	3	3	3

M = Massive

T/P = Tabular or Platy

I = Irregular

N = Narrow

I = Intermediate

T = Thick

VT = Very Thick

F = Flat

I = Intermediate

S = Steep

U = Uniform

G = Gradational

E = Erratic

Table 23.4.5. Ranking of Rock Mechanics Characteristics for Different Mining Methods

5b: Hanging Wall											
Mining Method	Rock Substance Strength			Fracture Spacing				Fracture Strength			
	W	M	S	VC	C	W	VW	W	M	S	
Open Pit Mining	3	4	4	2	3	4	4	2	3	4	
Block Caving Sublevel Stopping	4	2	1	3	4	3	0	4	2	0	
Sublevel Caving	-49	3	4	-49	0	1	4	0	2	4	
Longwall Mining	3	2	1	3	4	3	1	4	2	0	
Room and Pillar Mining	4	2	0	4	4	3	0	4	2	0	
Shrinkage Stopping	0	3	4	0	1	2	4	0	2	4	
Cut and Fill Stopping	4	2	1	4	4	3	0	4	2	0	
Top Slicing	3	2	2	3	3	2	2	4	3	2	
Square Set Stopping	4	2	1	3	3	3	0	4	2	0	
	3	2	2	3	3	2	2	4	3	2	

Key:

Rock Substance Strength

W = Weak

M = Moderate

S = Strong

Fracture Spacing

VC = Very Close

C = Close

W = Wide

VW = Very Wide

Fracture Strength

W = Weak

M = Moderate

S = Strong

5a: Ore Zone

Mining Method	Rock Substance Strength			Fracture Spacing				Fracture Strength		
	W	M	S	VC	C	W	VW	W	M	S
Open Pit Mining Block	3	4	4	2	3	4	4	2	3	4
Caving Sublevel	4	1	1	4	4	3	0	4	3	0
Stoping Sublevel	-49	3	4	0	0	1	4	0	2	4
Caving Longwall Mining	0	3	3	0	2	4	4	0	2	2
Room and Pillar Mining	4	1	0	4	4	0	0	4	3	0
Shrinkage Stoping	0	3	4	0	1	2	4	0	2	4
Cut and Fill	1	3	4	0	1	3	4	0	2	4
Stoping	3	2	2	3	3	2	2	3	3	2
Top Slicing	2	3	3	1	1	2	4	1	2	4
Square Set Stoping	4	1	1	4	4	2	1	4	3	2

5c: Footwall

Mining Method	Rock Substance Strength			Fracture Spacing				Fracture Strength		
	W	M	S	VC	C	W	VW	W	M	S
Open Pit Mining Block	3	4	4	2	3	4	4	2	3	4
Caving Sublevel	2	3	3	1	3	3	3	1	3	3
Stoping Sublevel	0	2	4	0	0	2	4	0	1	4
Caving Longwall Mining	0	2	4	0	1	3	4	0	2	4
Room and Pillar Mining	2	3	3	1	2	4	3	1	3	3
Shrinkage Stoping	0	2	4	0	1	3	3	0	3	3
Cut and Fill	2	3	3	2	3	3	2	2	2	3
Stoping	4	2	2	4	4	2	2	4	4	2
Top Slicing	2	3	3	1	3	3	3	1	2	3
Square Set Stoping	4	2	2	4	4	2	2	4	4	2

Table 23.4.6. Weighting Factors

Ore Geometry	1.0	1.0	1.0
Ore Zone Ground Conditions	1.33	0.75	1.0
Hanging Wall Ground Conditions	1.33	0.6	0.8
Footwall Ground Conditions	1.33	0.38	0.5

Table 23.4.8. Characteristic Values Multiplied by Weighting Factors

Mining Method	Geometry/Grade Distribution	Rock Mechanics Characteristics			Grand Total	
			HW	FW		Total
Open Pit Mining	12	9	8.8	4.0	21.8	33.8
Block Caving	11	8	4.8	3.5	16.3	27.3
Sublevel Stoping	11	5	5.6	1.0	11.6	22.6
Sublevel Caving	13	7	4.8	1.5	13.3	26.3
Longwall Mining	−37	8	4.0	3.0	15.0	−22.0
Room and Pillar Mining	−38	7	6.4	1.5	14.9	−23.1
Shrinkage Stoping	11	6	4.8	4.0	14.8	25.8
Cut and Fill Stoping	7	8	5.6	5.0	18.6	25.6
Top Slicing	14	6	4.8	3.5	14.3	28.3
Square Set Stoping	8	8	5.6	5.0	18.6	26.6

Table 23.4.7. Example of Numerical Model Selection Process

Geometry/Grade Distribution	(Column 1)	(Column 2) open pit (values from Table 23.4.4)	(Column 3) block caving etc. (values from Table 23.4.4)
General shape	tabular or platy	2	2
Ore thickness	thick	4	2
Ore plunge	flat	3	3
Grade distribution	uniform	3	4
Depth (used later)	425 ft (130 m)	--	--
		12	11
Rock Mechanics Characteristics		(values from Table 23.4.5)	
<i>Ore Zone</i>			
Rock substance strength	strong	4	1
Fracture spacing	close	2	4
Fracture strength	moderate	3	3
		9	8
<i>Hanging Wall</i>			
Rock substance strength	strong	4	1
Fracture spacing	wide	4	3
Fracture strength	moderate	3	2
		11	6
<i>Footwall</i>			
Rock substance strength	moderate	4	3
Fracture spacing	close	2	3
Fracture strength	weak	2	1
		8	7

Characteristic Values Totaled for Different Mining Methods

Mining Method	Geometry/Grade Distribution	Rock Mechanics Characteristics				Grand Total
		Ore	HW	FW	Total	
Open Pit Mining	12	9	11	8	28	
Block Caving	11	8	6	7	21	
Sublevel Stopping	11	5	7	2	14	
Sublevel Caving	13	7	6	3	16	
Longwall Mining	—37	8	5	6	19	
Room and Pillar Mining	—38	7	8	3	18	
Shrinkage Stopping	11	6	6	8	20	
Cut and Fill Stopping	7	8	7	10	25	
Top Slicing	14	6	6	7	19	
Square Set Stopping	8	8	7	10	25	

Table 23.4.9. Summary of Methods Selected

	Boshkov and Wright	Hartman	Morrison	Laubscher	Nicholas
Open Pit Mining		X			1
Block Caving	X	X	X	X	3
Sublevel Caving	X	X	X		5
Top Slicing	X		X		2
Square Set Stoping	X				4
Cut and Fill Stoping					6

On the basis of relative operating costs, the methods would be ranked as follows:

- 1. Open pit mining.**
- 2. Block caving.**
- 3. Sublevel caving.**
- 4. Top slicing.**
- 5. Square set stoping.**

Table 23.4.10. Ranking of Mining Methods Based on Relative Operating Costs

Mining Method	Ranking Cost (Percent)
a. Hartman (1987) Ranking	
Hydraulicking, Dredging, Leaching	5
Open Pit/Open Cast Mining	10
Block Caving, Longwall Mining	20
Room and Pillar Mining	30
Stope and Pillar Mining	
Sublevel Stoping	40
Shrinkage Stoping, Sublevel Caving, Induced Caving	50
Cut and Fill Stoping	60
Square Set Stoping	100
<i>Source: Hartman, 1987.</i>	
b. Morrison (1976) Ranking	
Open Pit Mining	Lowest
Block Caving	
Sublevel Stoping	
Sublevel Caving	
Longwall Mining	
Room and Pillar Mining	
Shrinkage Stoping	
Cut and Fill Stoping	
Top Slicing	
Square Set Stoping	Highest

STAGES IN THE SELECTION PROCESS

The mining method selection process is iterative, with each iteration correlating to the stages of development of the property. A deposit that is mined will go through roughly five stages:

- ▶ Initial discovery of the mineralized zone.
- ▶ Developmental drilling to outline reserves and a feasibility study.
- ▶ Design and development of the deposit.
- ▶ Revising and updating during production.
- ▶ End of mine life.

Stope design and computer application

- ▶ Stope is an ore block of proper size prepared out of a deposit to be mined, for the purpose of final exploitation of the ore body enclosed within it.
- ▶ Stope dimensions differ from method to method and even for the same stoping method from mine to mine.

(While deciding stope dimensions strength of orebody and its enclosing walls, rate of stoping, degree of mechanization, depth, size and shape of the orebody play important role)

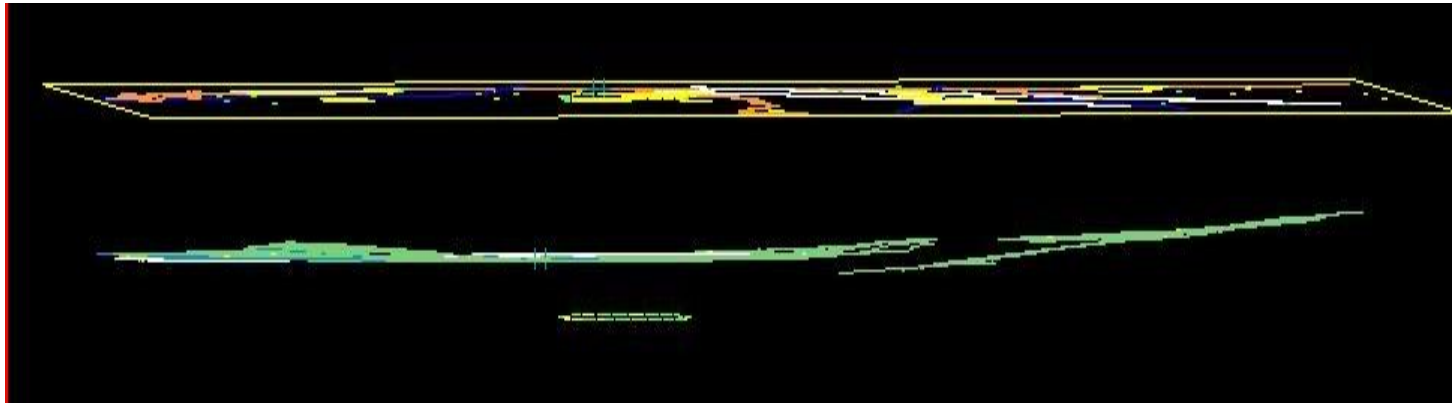
- ▶ In absence of adequate geo-mechanical studies to know the ground stability pattern of a stope during its stoping operations, the experience often play an important role to decide its size. And, therefore, usually the dimensions known to be safe are adopted.
- ▶ Level interval usually decides the stope height, its width remains to be that of the orebody width, and the length is decided considering the ground stability parameters

Based on these considerations while designing the stope following parameters are laid out:

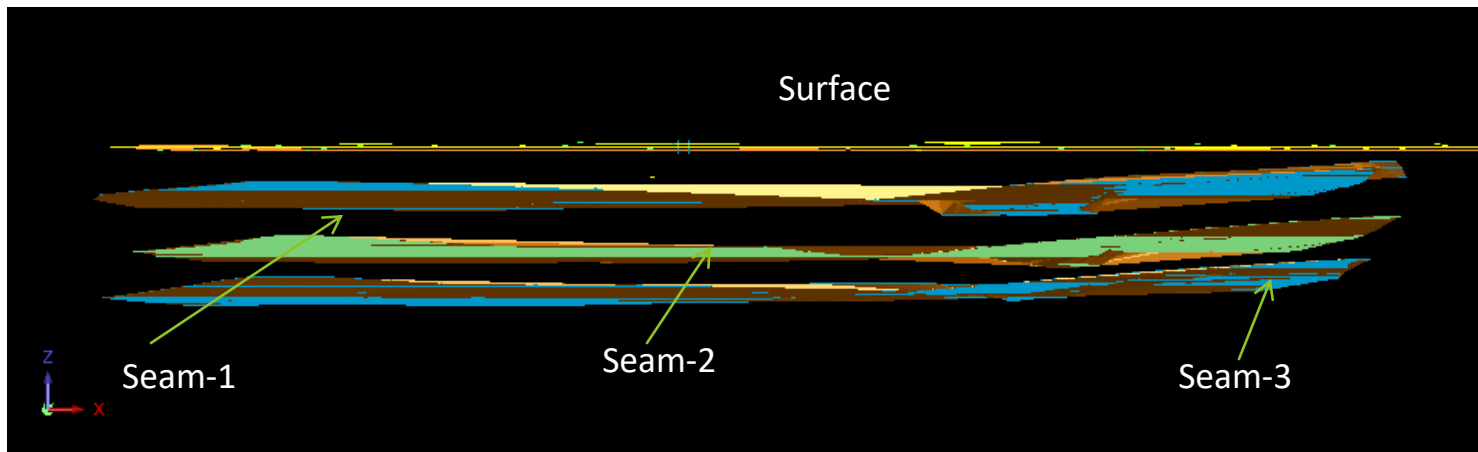
- ▶ Model parameters
- ▶ Design parameters

Model parameters

This includes the parameters such as orebody profile in plan (at various horizons) and sections (along various latitudes and departures) which, in turn, depicts orebody's thickness and dip.

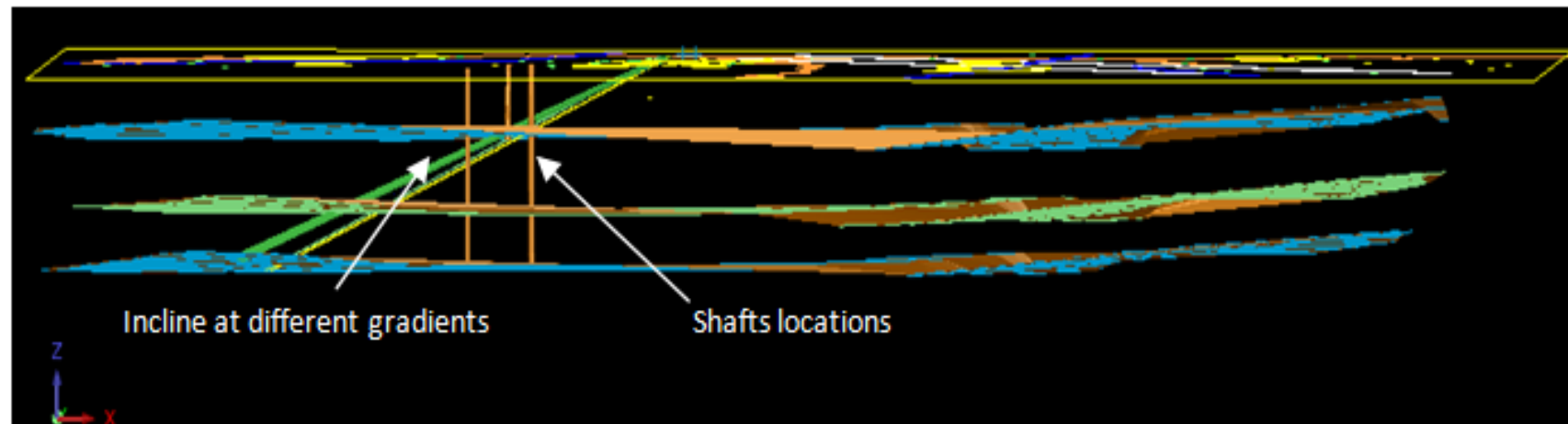
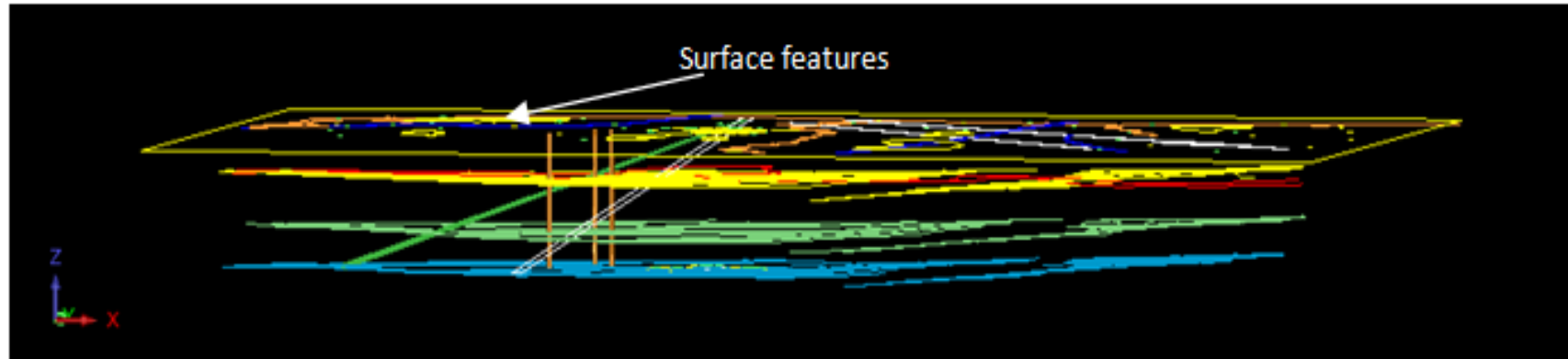


 3D- model of digitized surface and seam-1 contour plans
prism4.dtm



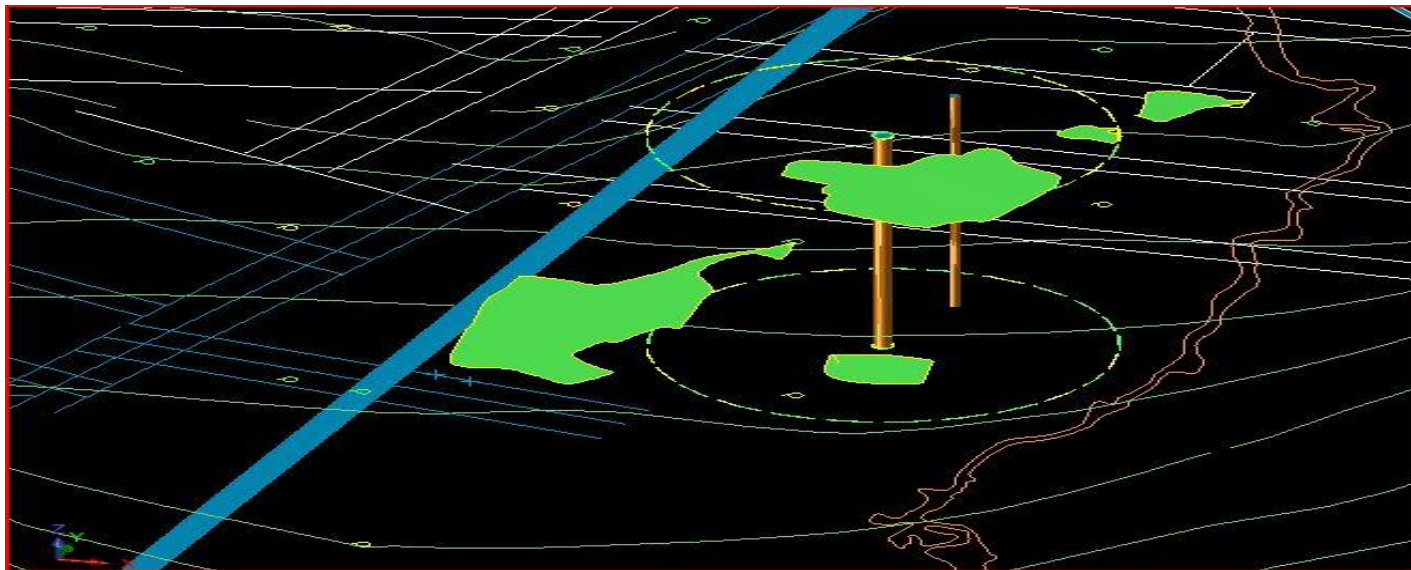
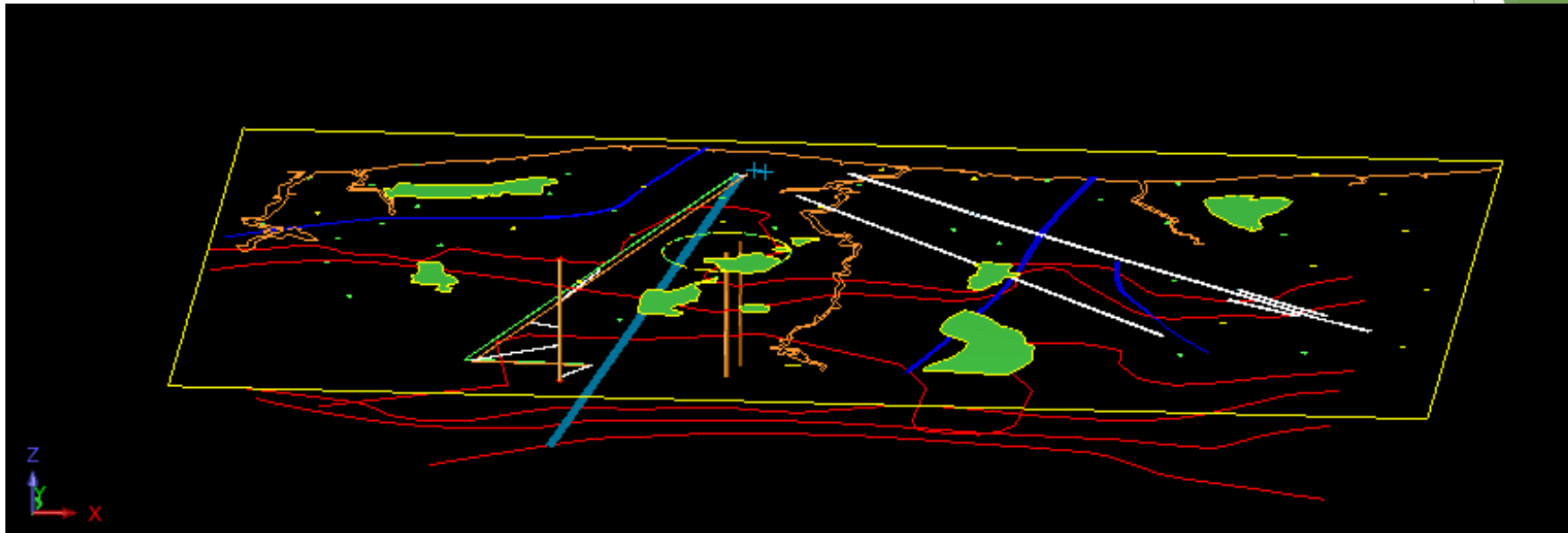
Composite 3D- model of digitized surface contours and
seams

Design Approach

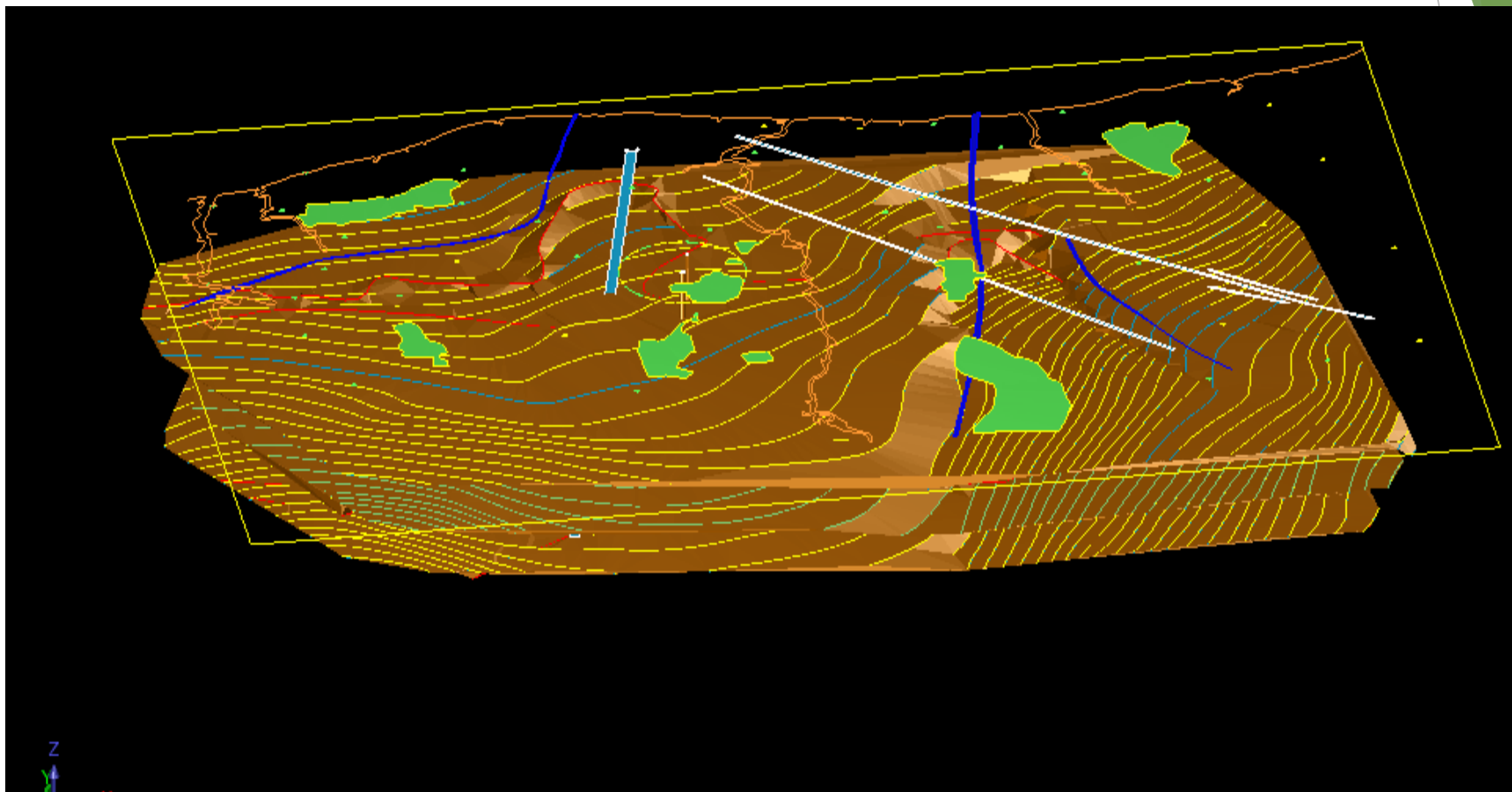


Study of various options for mine design

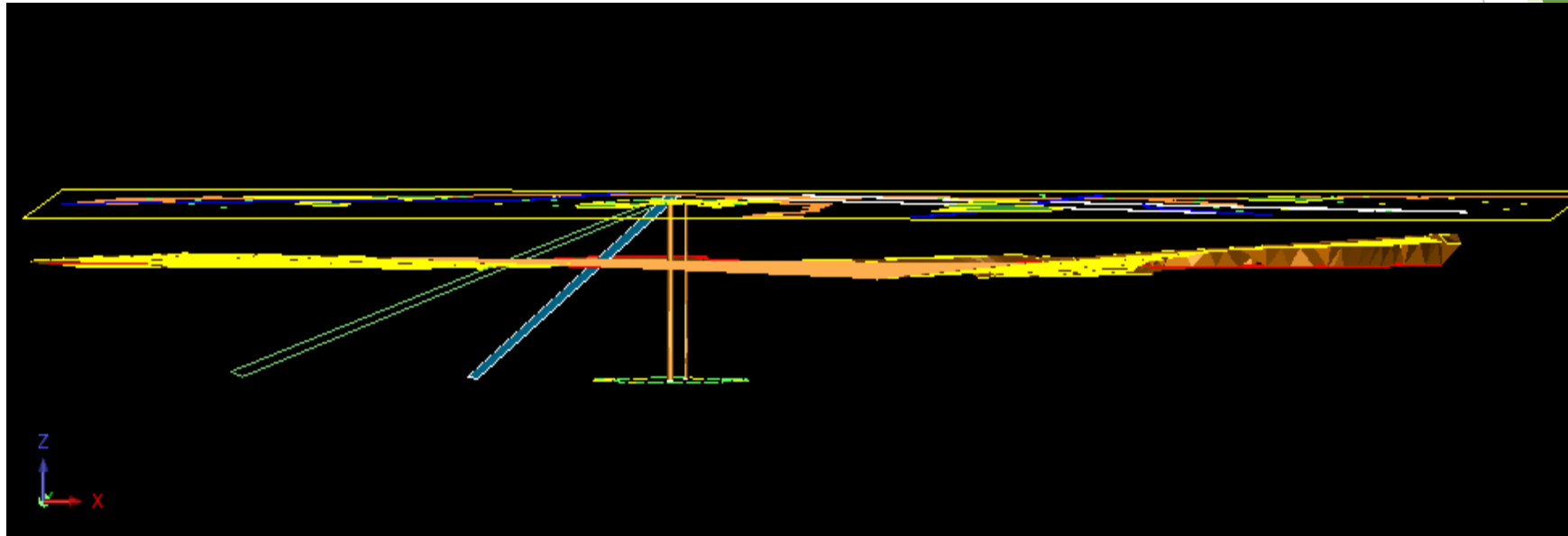
Mine design exercises



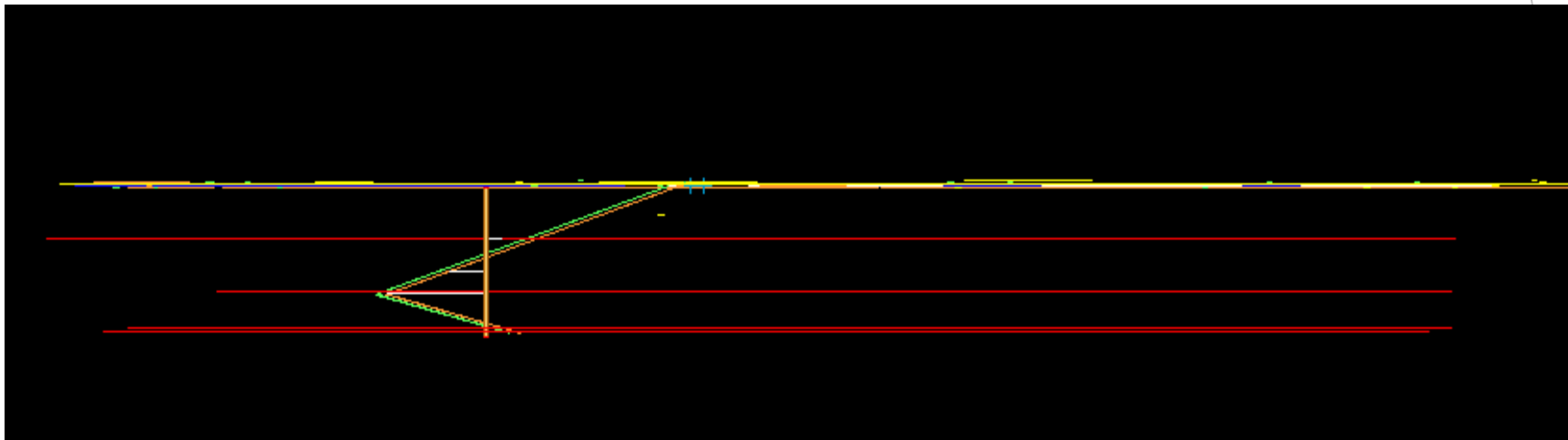
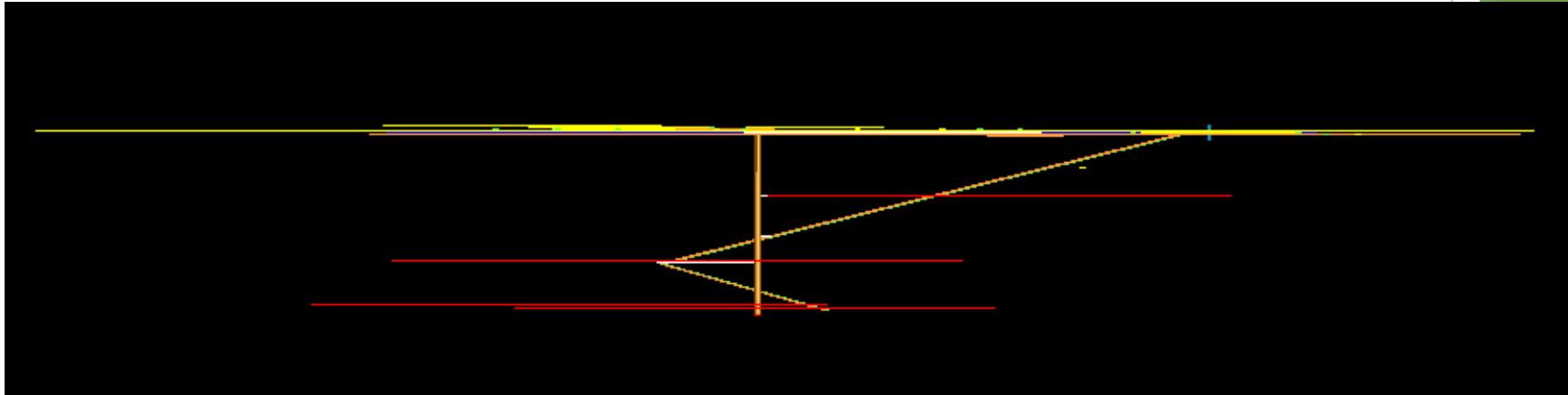
Mine design exercises with DTM of different seams



Design Approach



Final design after optimized



Design parameters

In order to mine out a stope, following provisions need to be made:

- ▶ Access to the stope for workers, machines, equipment and mine services (Service Raise or Manpass).
- ▶ Creation of free face for the stoping operations to start with (Slot).
- ▶ Rock fragmentation by way of drilling and blasting the blast holes (Drill drives/sublevels or cross cuts).
- ▶ Handling ore fragmented muck (Extraction level layout).

Trough, Funnel ,Grizzly , Finger raise system ,Ore pass system Scraper

- ▶ Limiting and isolation of stope size or dimensions (Pillars of various kinds).

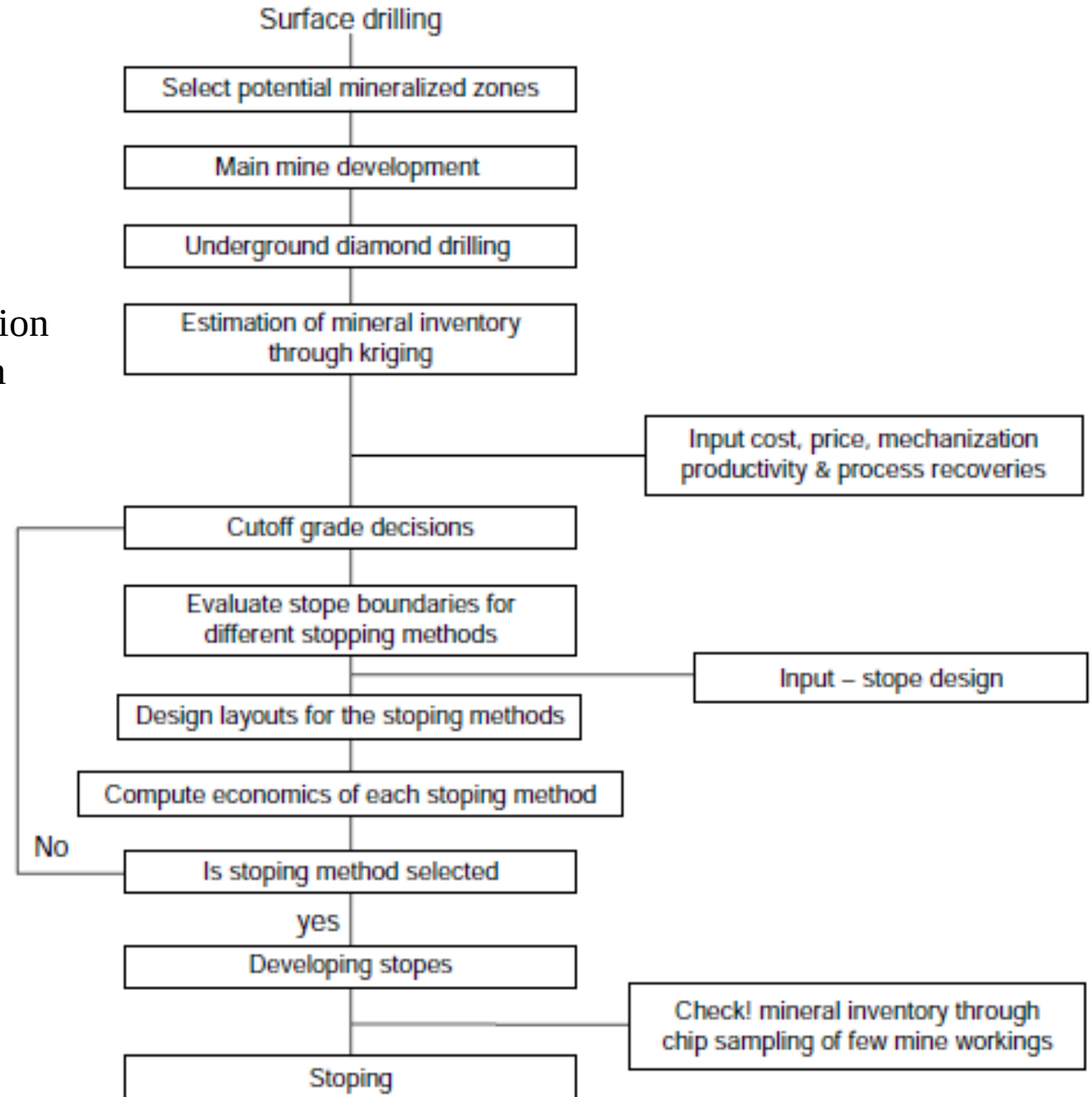
APPLICATION OF COMPUTERS IN STOPE DESIGN AND ECONOMIC ANALYSIS

SELECTION OF A STOPPING METHOD

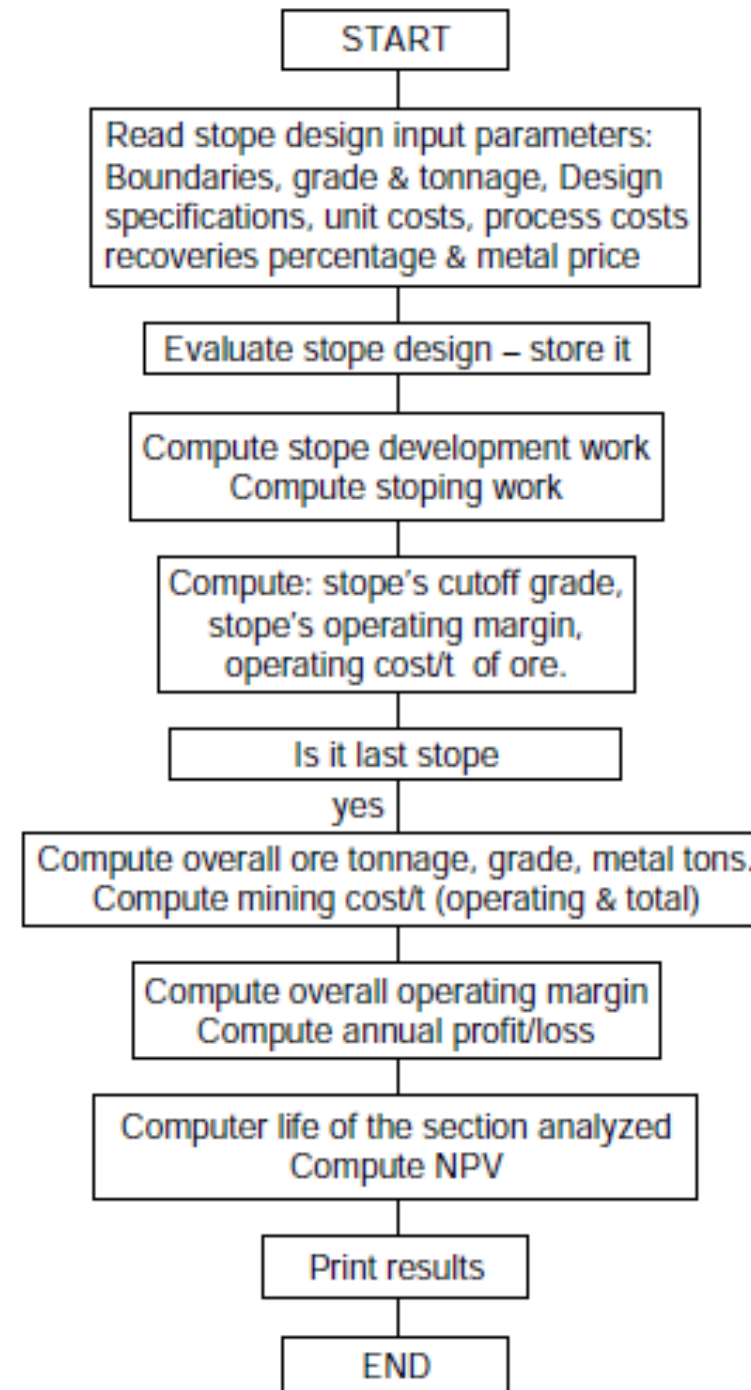
Flow diagram for the methodology to select stoping method for a base metal deposit

This can be interpreted as aiming for maximum extraction with safe working conditions at low cost and maximum productivity.

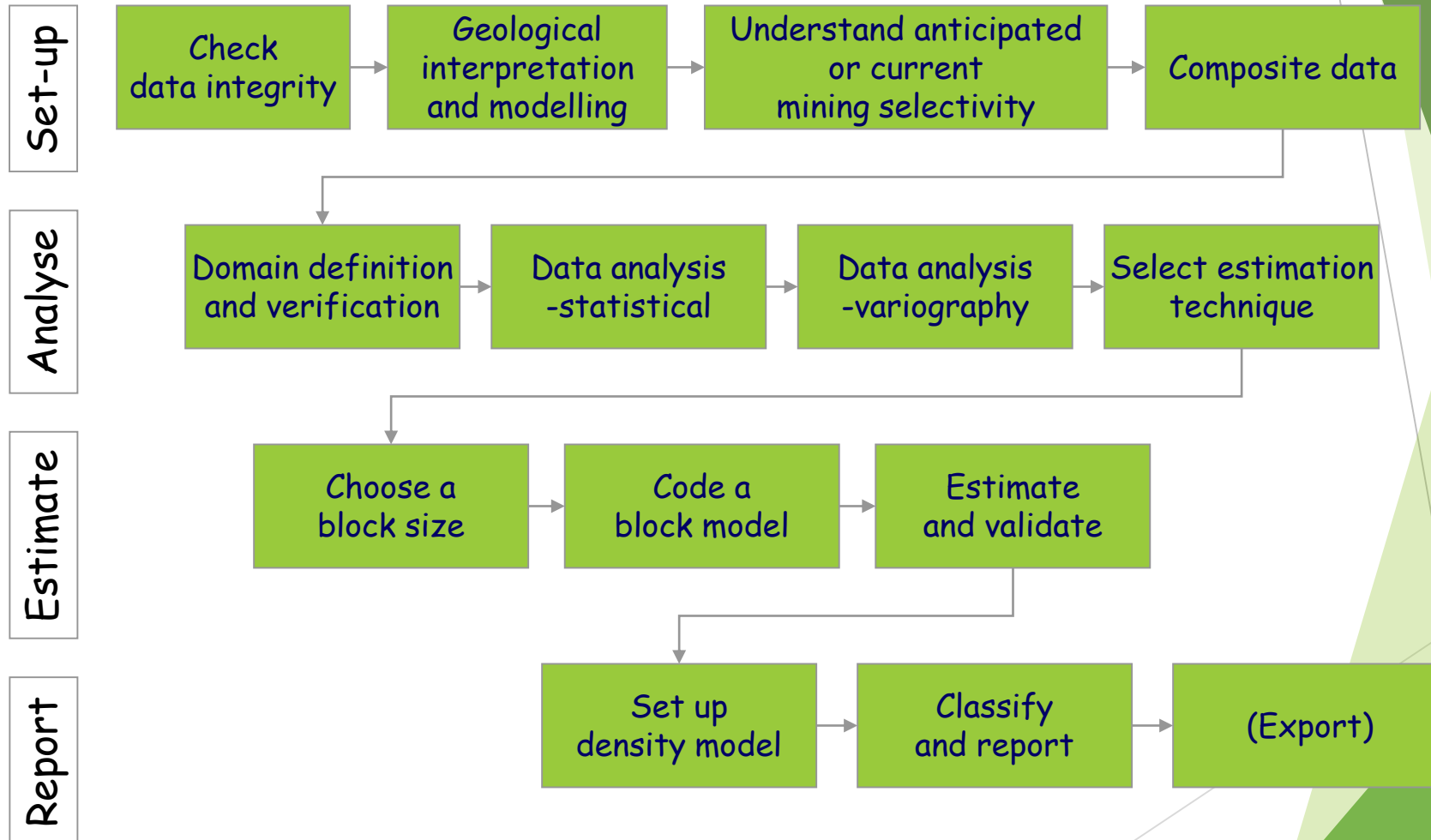
- *Estimating mineral inventory from the basic geological information.*
- *Formulating a basis for cutoff grade decisions to predict mineable ore reserves.*
- *Evaluating stope boundaries for different stoping methods.*
- *Designing stopes and their economic evaluation for the different mining methods through the algorithms developed.*
- *Selection of stoping method.*



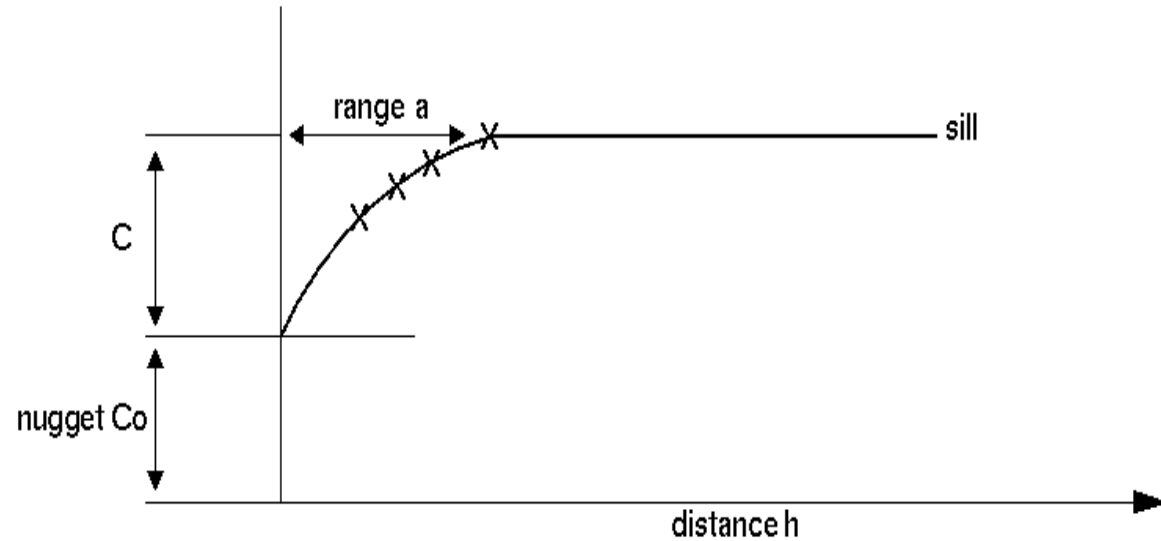
Flow diagram for computer assisted stope design and economic analysis



The resource modelling process



Estimating mineral inventory from the basic geological information



Mathematically it is represented as :

$$\gamma(h) = (1/2n) \sum [Z(x+h) - Z(x)]^2$$

Where n = Number of sample pairs at distance h .

h = lag distance

$Z(x)$ = Sample value at point

' x '.

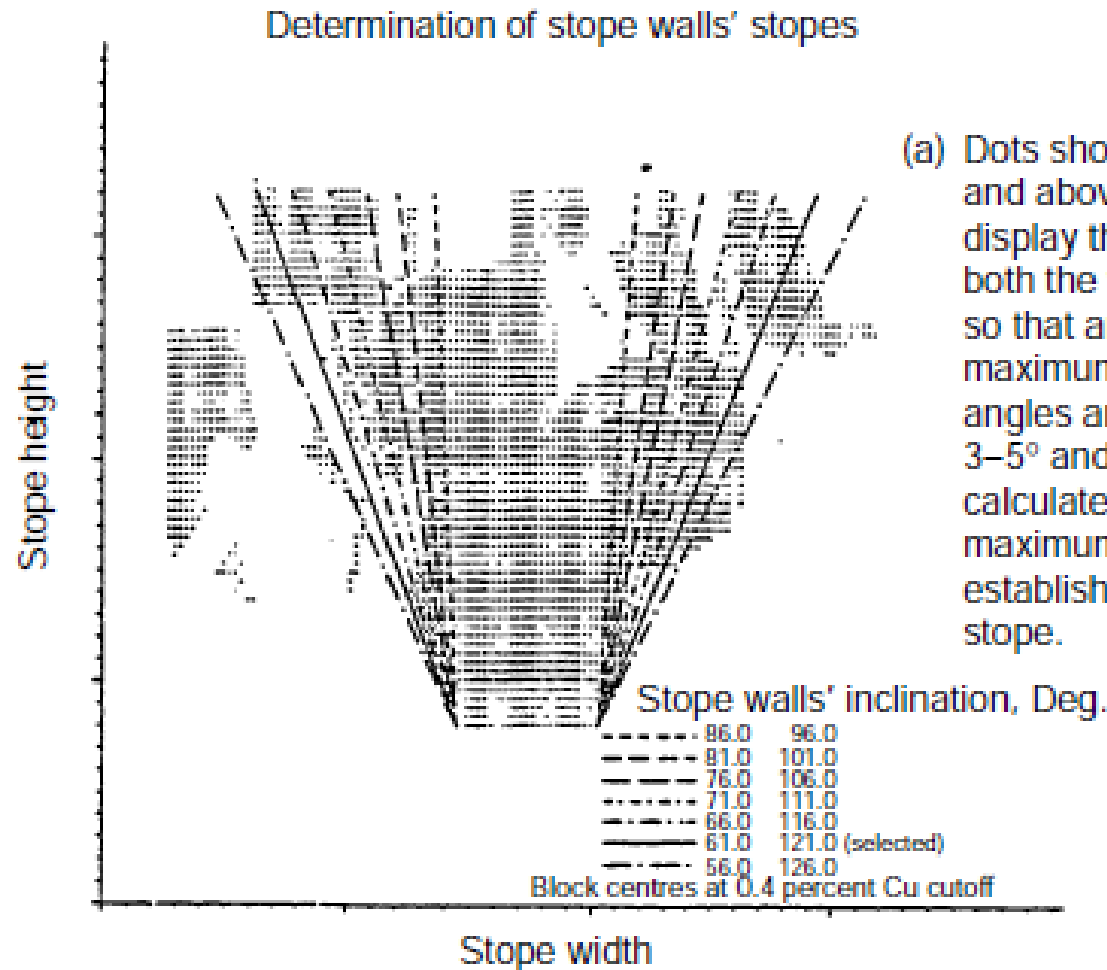
$Z(x+h)$ = Sample value at point

' $x+h$ '

Variogram is a Mathematical function/tool to represent the spatial continuity of basis data set. In other words it may be represented as a statistically based, quantitative description of surface's roughness.

Accurate variogram modeling is the prerequisite to any geostatistical study as this is the most basic function which controls the subsequent estimators.

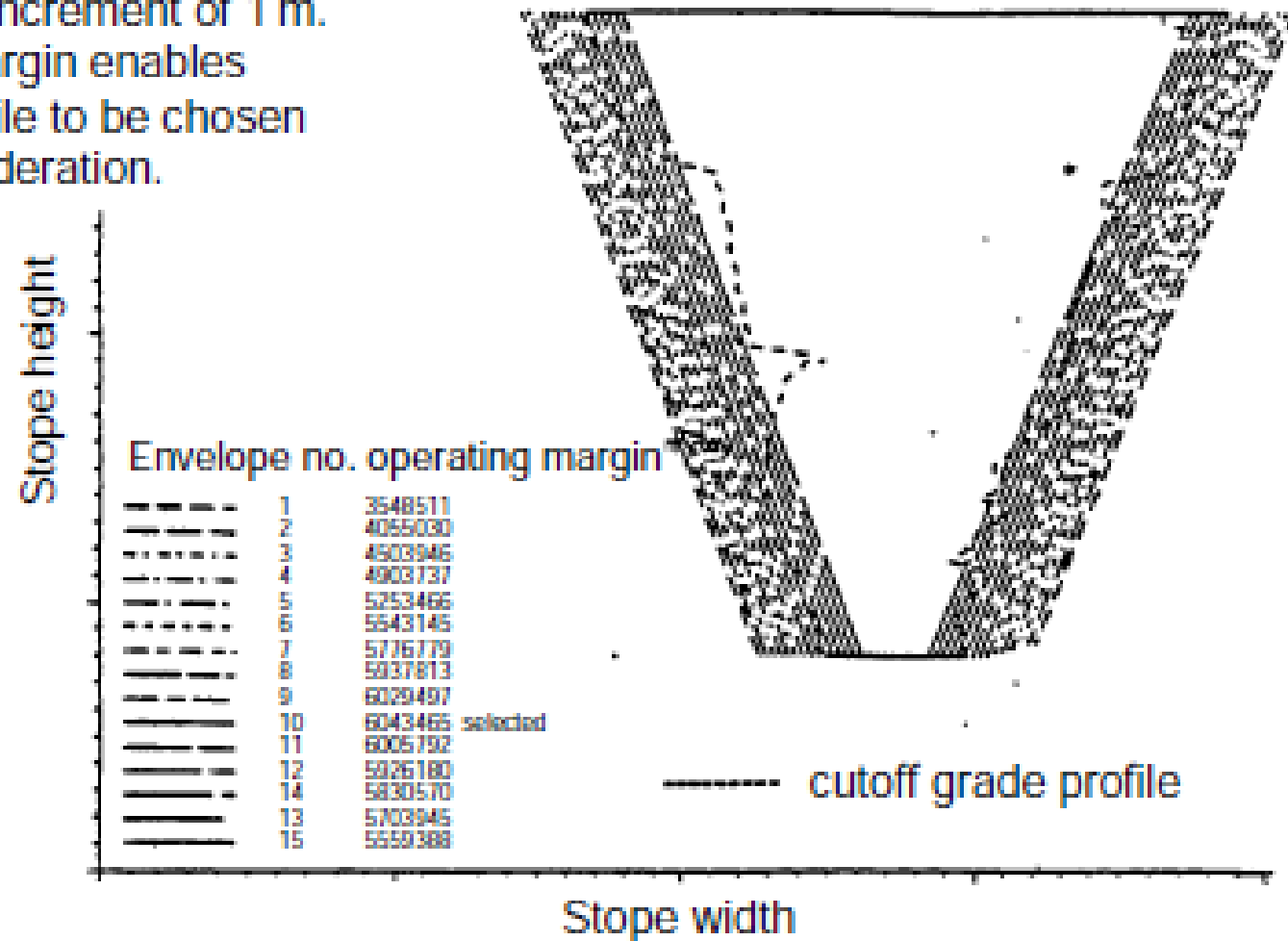
Evaluating stope boundaries for different stoping methods using cutoff grade and incremental analysis criteria.



- (a) Dots shown are the blocks that are at and above cutoff grade. From this display the wall angles are then for both the walls are selected manually so that angle chosen should include maximum number of blocks. The wall angles are increases incrementally, 3–5° and economics of each envelop calculated. That which yields the maximum operating margin thus establishes the wall angles of the stope.

- (b) The ultimate stope profile is then derived by moving horizontally the stope walls by parallel increment of 1 m. Maximum operating margin enables the optimum stope profile to be chosen for a stope under consideration.

Determination of stope walls by incrementing strips.



- In underground situation this is interpreted as a number of ‘envelopes’ that could form possible stope boundaries. An envelope is defined as the mineralized zone bounded by the upper and lower development levels (i.e. level interval), the interface with the adjacent stope and the two side walls. Price of metal, revenues from the byproducts, ore and metallurgical recoveries, total variable costs mining through to the process plants, royalty, excise duty and selling costs have been used to compute the operating margin of an envelope.

A computer printout showing the activities and cost analysis for stope development and stoping for a DTH stope

Operation	Quantum of work	cost/unit in Rs*	Total cost in Rs*
Development (horizontal)			
Extraction level, cu.m	5059.45	263.57	
Drilling level, cu.m	5655.64	263.57	
Crown sublevel, cu.m	676.48	263.57	
Decline (common), cu.m	122.50	263.57	
Total horizontal development, cu.m	11514.07		#
Total vertical development, cu.m	556.53	417.32	#
Longhole drilling, m	20931.61	102.14	#
DTH drilling (165 mm ϕ)	3952.01	235.71	#
Stoping (blasting, mucking & trans.)	243760.21	26.81	#
Hoisting tonnes	279972.00	7.50	#
Miscellaneous tonnes	279972.00	5.35	#
Total variable cost			#
Variable mining cost/t			#
Cutoff grade			0.352

* Currency units which could be as applicable to the country of application.

Calculation for this column for the corresponding parameters proceeds but not shown here.

A computer printout showing the techno-economical details for a DTH stope.

Parameters	Quantum
Percentage extraction	88
Stope reserves in tonnes	279972.00
Average grade in % Cu	1.10
Quantum of metal in tonnes	2556.23
Total revenues in Rs*	#
Total cost of mining in Rs*	#
Operating margin in Rs*	#

* Currency units which could be as applicable to the country of application.
Calculation for this column for the corresponding parameters proceeds but not shown here.

Comparison of the techno-economics of the section analyzed for the three stopping methods

Techno-economical parameters	Sublevel stoping	DTH stoping	Cut and fill stoping
	(Alternative-1)	(Alternative-2)	(Alternative-3)
Section's total reserves in million tons.	1.386	1.586	1.322
Average grade in % Cu	0.75	0.70	0.76
Cutoff grade in % Cu	0.43	0.35	0.51
Metal recovery in tons.	8736	9466	8616
Operating margin in millions Rs*./yr. (considering variable mining costs only)	278.86	289.52	258.90
Operating margin in millions Rs*./yr. (considering total mining costs only)	216.06	226.72	196.10
Operating margin in millions Rs*./yr. (considering mining costs mining through process plants)	-248.44	-217.75	-276.58
NPV of the section analyzed (in million Rs*.)	-209.51	-208.75	-222.83
Life of the section in years	0.924	1.057	0.881

* Currency units which could be as applicable to the country of application.

- ▶ This methodology, in general, provides a basis to select the mineral inventory, estimated through geostatistical or any other technique, for base metal deposits suitable for mining.
- ▶ Using this methodology the internal waste or sub-grade material contained within a stope can be assessed. In addition, it provides a basis to compare the methods that could be technically feasible to mine out the same chunk of a deposit.
- ▶ The computer assisted stope design has automated many of the manual procedures and calculations involved in mine planning. This can allow evaluation of many layouts and mining strategies